

Aptiv 2023 | Aptiv and Wind River Software Teach-in

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OPENING REMARKS

Jane Wu | Vice President, Investor Relations and Corporate Development

Welcome, and thank you for joining Aptiv and Wind River's Software Teach-In. We hope that today's event provides an engaging and interactive opportunity for you to learn more about our software strategy and how we are enabling a software-defined future.

Following our presentation, which features a number of our business and technology leaders, I will moderate a live Q&A session. Questions can be submitted through the Slido text box below your video window at any time during today's event, which is expected to run approximately 75 minutes.

Before we begin, I would like to remind you that today's discussion will include forward-looking statements based on our current view of future financial performance, and that actual results may differ for reasons that we cite in our form 10-K and other SEC filings. We will not be addressing specific questions about the quarter or year, and we will defer any financial questions until our third quarter 2023 earnings conference call. A replay and transcript of today's event will be published on our website at ir.apativ.com.

With that, let's get started.

WELCOME

Kevin Clark | Chairman and Chief Executive Officer

Hello everyone, and welcome to the Aptiv-Wind River Software Teach-In. We are really excited to provide you with an update on our software strategy.

As you are aware, the digital transformation of the industrial landscape is well underway. The devices around us continue to get smarter and generate more data, requiring more processing at the edge and higher levels of compute performance.

To keep up with the pace of change, these smarter devices require cloud-native, intelligent edge software solutions that deliver the highest levels of security, safety, reliability and performance.

As a global leader in intelligent edge software, Wind River has helped enable the ongoing digital transformation of a variety of industries, including Aerospace and Defense and Telecom. In fact, Wind River software solutions are deployed on over 2 billion devices worldwide, most of which require secure development, deployment, operation and servicing of mission-critical software.

Now, to enable the software-defined vehicle of the future, the automotive industry needs to fully embrace the digital transformation that is taking place in these other industries, and the combination of Aptiv and Wind River are perfectly positioned to drive this change.

Aptiv's compute, application software, and vehicle architecture expertise, combined with Wind River's full edge-to-cloud software portfolio, delivers significant benefits to our OEM customers, including reducing cost and complexity, increasing flexibility and unlocking new business models.

At Aptiv, we continue to solve our industries' toughest challenges by providing high value, margin accretive solutions which create value for our shareholders.

Today you will hear from some of our top software and technical talent – Benjamin Lyon and Anant Thaker of Aptiv, as well as Paul Miller and Avijit Sinha of Wind River – and there is a lot to talk about, so let's get started.

I will now turn it over to Benjamin Lyon, Aptiv's Chief Technology Officer, who will kick off today's discussion.

INDUSTRY TRENDS

Benjamin Lyon | Senior Vice President and Chief Technology Officer

Thanks Kevin.

In recent years, the landscape of business and consumer interactions has been revolutionized by the ongoing digital transformation.

As Kevin mentioned, devices around us continue to get smarter and need to handle massive amounts of data. Consumers are now demanding that things around us are continuously connected and kept up to date. Just think about how we interact with our smart phones and the internet every single day.

To unlock the full potential of this digital transformation, we must address the technical challenges that lie ahead and demand continuous innovation, with a focus on enhancing processing power, upgrading software capabilities and enabling true edge-to-cloud connectivity.

Today, the industries undergoing the digital transformation are coping with a shared set of challenges – increased software complexity, a lack of modern tools, difficulty in balancing connectivity and security and proliferation of data to just name a few.

There are three key architectural puzzle pieces that we are bringing together to solve these challenges, creating a blueprint applicable across multiple industries. Let's start at the foundation of the technology stack and work our way up.

First we have next-gen hardware, which is designed for mixed-criticality workloads and enables an up-integrated and virtualized compute platform.

Moving up the stack, we are enabling cloud-native software architectures. This modular approach leverages microservices and containers to build, run and update software. This provides significant advantages versus a monolithic architecture.

And finally, we are providing an edge-to-cloud connectivity platform. This end-to-end, cloud-native platform streamlines the entire software development lifecycle. This enables seamless and secure development, deployment and operation of complex software ecosystems.

By adopting these solutions, our customers can not only navigate the complexities of transitioning to a software-defined future, but also lower cost and pave the way for future growth in their own software offerings.

At the forefront of the next wave of digital transformation are four industries, characterized by their complex operations and intricate networks: Telecommunications, Aerospace and Defense, Automotive and Industrial.

The concept of the intelligent edge is central to this transformation, enabling the processing of larger data volumes and facilitating quicker, informed decisions at the source, fostering agility and responsiveness at the edge.

Predictions indicate an acceleration in the pace of transformation over the next two years. By 2025, a significant shift is expected, with 75% of edge-collected data being processed at the edge.

As this common trend spreads across these industries, it will inevitably create substantial business opportunities. Aptiv and Wind River's capabilities are perfectly situated to deliver the comprehensive solutions and unique tailoring each industry demands. As such, we are positioned to continue to capture substantial value from this transition.

While these industries navigate a similar path, their speed of transformation varies.

Leading is consumer electronics. Products like your smart phone and edge-enabled features like video meetings are examples of mature deployments of cloud-native, edge enabled software-defined architectures. So we see in consumer products all three architectural puzzle pieces developed and deployed at scale.

Next is the telecommunications market, where this architecture is in initial production and is in the process of scaling up. And it is followed by other industries, including aerospace and defense, automotive and industrial, each at varying levels of maturity.

For automotive, the industry is rapidly moving towards next-gen architectures, and Aptiv is leading this charge with our Smart Vehicle Architecture™ solutions. We'll touch on that in more detail later on.

As industries continue along this transformation, there is a significant market opportunity for edge-to-cloud software solutions, underscored by the projected growth across these end-markets. By 2030, we estimate a total addressable market of \$21 billion, indicative of the need for mission-critical software across multiple evolving industries.

Let's drill into the double-digit growth drivers of each.

In telecommunications, the market is scaling to cloud-native, software-defined infrastructure to support the roll out of 5G, a marked shift from proprietary bare metal solutions.

In aerospace and defense, which shares many of the highly regulated, safety critical requirements in automotive, the sector is urgently focusing on software to make quick decisions in high-risk scenarios, expanding the need for intelligent systems.

Such urgency is driving a similar large market opportunity in the automotive space, as the industry needs to transform from legacy to cloud-native architectures, to enable the software-defined vehicles of the future.

Similar attractive growth opportunities exist in the industrial market, as manufacturing becomes increasingly automated and software driven.

To capture the value from this tremendous market opportunity, Aptiv and Wind River will enable the transition from legacy to cloud-native software architectures.

Revisiting the three key puzzle pieces I mentioned earlier, let's start with next-gen hardware architectures. The industries we just discussed are all transitioning from single-use, distributed hardware, to up-integrated high-performance compute.

Up-integrated compute is needed to reduce overall hardware complexity and cost as well as break the dependency of the software on the underlying silicon. This allows greater flexibility in managing your silicon choices. It also decouples the software development process from the hardware development process. This provides scalability as new features are continuously added within these software architectures.

Moving up the stack, the application software in legacy architectures is monolithic and laced with complex interdependencies that slow down the development, testing and deployment of new features and applications.

Future software applications need to be architected with a modern, service-based approach that leverages standard containers. Why standard containers? Well containers help us break down that large, monolithic code into isolated, testable, scalable components that allow us to develop, deploy, operate and service software much more efficiently.

To take advantage of modern software architectures, you need a modern, unified, end-to-end connectivity platform that enables faster and more efficient development, deployment and operation across the full software lifecycle.

And finally, moving to cloud-native architectures will require the highest levels of safety and certification for use in the mission critical industries we just walked through.

Let's go one level deeper on mission critical systems. By fully addressing the mission critical nature of next-gen software architectures, Aptiv and Wind River can continue to provide more flexibility and better performance while reducing cost.

The up-integrated, virtualized hardware must support simultaneous execution of mixed-critical workloads, leading to a further centralization of software functions that reduces system complexity and costs.

The software applications ensure deterministic execution, assuring that the system performs desired actions within specified time constraints, consistently. This is especially important in aerospace and defense, as well as automotive, where a single software failure could have fatal consequences and lead to significant damages. For example, cybersecurity threats are constantly evolving, and the ability to address them efficiently and in a timely manner is absolutely essential in these industries.

And lastly, the edge-to-cloud platform must enable continuous certification, allowing software to be safely certified throughout the development process. This reduces the certification time and the costs associated with traditional approaches.

Now that we've walked through the broader industry transformation, I'd like to introduce you to Paul Miller, Wind River's Chief Technology Officer. Paul will discuss the Wind River portfolio and how they've solved some of these challenges across multiple industries. Paul?

WIND RIVER SOLUTIONS

Paul Miller | Wind River's Chief Technology Officer

My name is Paul Miller. I'm the Chief Technology Officer for Wind River, and I'm here to give you a bit of an overview of what we do at the company.

Wind River's history is one of being in mission-critical software systems used in aircraft, defense, automotive, telecom, medical, a whole host of areas where you'll find Wind River is supporting human lives when they're at risk. In particular, we provide ultra-reliable, safety-critical and high-availability software.

We have over 40 years of business of doing this. And in recent years we've produced a portfolio that's comprised of edge assets, as we call them, that you see in the bottom of the slide here. And these are really infrastructure components that are used to power these mission critical systems, whether they be aircraft or automobiles or medical devices.

In recent years, Studio has been introduced, as you see in the top right, and this is really growing the company from a software vendor to more of a strategic partner with our customers, and we help them solve software lifecycle and operational needs through this.

As we start looking at how people use this software, we're going to start with the edge assets in the portfolio. You'll see here on the left hand side, a layered diagram, which typically has some type of hardware platform that the software is deployed on. You can see Helix, VxWorks, and Linux sitting on that hardware. And then applications that our customers build are then deployed on that system.

This comprehensive portfolio gives us the ability to solve problems for our customers. And today, we're in over 2 billion devices providing those capabilities.

The variety of software that you see here includes Linux that's available for high performance embedded systems. VxWorks, which allows critical systems to be built in safety critical environments and safety certified environments. And Helix platform, which allows us to up integrate compute systems.

This has been used in avionics platforms for over 10 years, and this is providing the ability to have incredible savings in platforms where historically you may have tens or hundreds of computers with all the harness and weight that goes with that type of a distributed system. Now, being up-integrated via software to produce massive savings in these systems, we're really unique in the market in supporting modern cloud-native and containerized applications.

If you look at the history of embedded systems, they were really monolithic pieces of software that just ran directly on bare metal. As modern technologies for software development have moved forward, people are now building software with containerized capabilities.

This provides a much more flexible software development environment, a much faster time to market and significant savings in the effort required to develop that software. This is a very attractive way to build software in today's modern world, and we're bringing that to the embedded device portfolio as well.

So as we move from the edge portfolio over to studio, you can see this diagram in the bottom center, and this is known in the industry as DevSecOps. This is the software lifecycle methodology that people follow to build software and maintain it as a complete lifecycle operation.

It starts in the Plan phase where you ideate the concept of a product. As that moves into the Code or software development phase, people actually build the software that's used to implement the product. You then move towards Build where a binary is created and delivered into that system. That system then goes through significant test activity to validate the functionality is as intended. Then the output of this is in the Release phase where a software artifact is produced that is then used in the system.

For many businesses, they spend a significant amount of investment in their software development and have difficulties hitting quality software, maintaining that software in the field and hitting their schedules.

This tool chain that we've developed, Studio Developer, solves this particular phase of the DevSecOps problem. It's the industry's only cloud-based development solution enabling significant savings in OPEX, as well as improved time to market and maintainability through having a modern tool chain.

As you can see on the top element of the slide, the various components that we provide in this solution.

You can see the Studio Gallery where we provide the ability to bring third party tools integrated into the tool chain.

Pipeline Manager they can visualize their entire development cycle through this DevOps loop and interact with the various tools that they need through a cloud-based single pane of glass.

The Test Automation framework provides the ability to fully test the software that they're building in a virtualized environment, in a fully automated fashion, integrated with continuous integration and continuous deployment.

The Virtual Lab ability provides a virtual target that emulates their actual system to test in a cloud-based environment as if they were testing on the actual physical device.

And then System Simulation provides the ability to shift left software development instead of waiting for hardware prototypes to arrive with new silicon. Systems can be simulated and software can be built and run in those systems before they even physically exist. This provides tremendous time to market advantages for customers as they use this tool chain.

As we shift now to the Operations side, we'll take a look at the concept of Deploy and Operate.

Once the Developed capability is complete, you now have produced a software artifact. Handoff from Developer occurs with a software artifact produced from the development tool chain being deployed into a system. This can occur at the time of manufacturing or in the field, depending on the industry.

The Operate capabilities then enter. This provides the ability to upgrade, patch and provide security updates as well as lifecycle manage that deployed system. This is a very modern capability. You can see it's a continuous loop where the results of that Operate cycle are continuously iterated back into the development tool to ensure you have a continuous integration and continuous deployment capability that allows you to continuously adjust and improve the software even though it's deployed in those remote systems

And then finally, the monitor capability within the Operate quadrant. We call this our Digital Feedback Loop. This collects data and it feeds the digital twin or development updates and produces live analytics from deployed systems.

As you see the components here, we have Cloud Platform, which is an advanced Kubernetes system that allows a fully cloud managed environment for distributed systems.

We have the Digital feedback loop capability that allows collection from the field data and informing the backend development and digital twin.

And then Wind River Conductor that provides orchestration and automation for these deployed systems

As well as analytics that collects data from the field and presents that through visualizations and is used to feed machine learning models.

As we move to a few industry examples, we'll start with aerospace and defense. Wind River's been in aerospace and defense for almost 40 years, and our technology is used in avionics systems, critical defense infrastructure, land, sea, air and cyber assets in the US and internationally.

A few different programs going on now that are very interesting and relevant to our technologies. First, with Studio Operator, we're involved in what's called JADC2 and Platform One. JADC2, it's a distributed cloud technology that allow the different service branches to operationally manage hugely distributed tactical assets, whether they're aircraft or ground-based vehicles or ships, and tie them together in a modern software paradigm. Platform One is really fed by Studio Developer that enables us to bring at DevSecOps tool chain for the development of the software assets that are then deployed into the field.

You can see here in the aerospace and defense industry many parallels with what's starting to happen in the automotive industry as the automotive industry looks to provide over-the-air updates and manage a software-defined vehicle.

There's tremendous benefits of this cloud native architecture to aerospace. It provides modularization, reduced costs, the ability to re-use software components, and it reduces complexity through that modularity because the software's being broken down into individual containerized components that can be separately developed with standard APIs. As a result of this, you achieve improved reliability and reduced time and development.

At the bottom here you can see an interesting example. Using our Helix hypervisor, we were able to up integrate a variety of hardware platforms into a single, virtualized infrastructure that allowed us to have a single compute platform that virtually hosted the multiple separate functions that used to be in the aircraft previously.

This provides weight savings and miles of wire harness elimination, reduces cost in the airframe and creates lifetime fuel savings from the weight savings that happens from the reduced compute and harness workload in that vehicle. Aircraft have used this approach to up-integrate over 200 separate functions into a single computer in this industry using Wind River's technology previously.

As we then take a look at the telecommunications industry, there's been a major transformation happening in this industry over the last several years. It's really transformed into a software-defined, cloudified infrastructure. This started at the core of the network and it has now moved all the way to the edge where 5G is being deployed today.

This has resulted for those service providers lower cost and faster time to market as they can deploy software functions with the click of a button, rather than taking years to integrate new hardware platforms. This has also eliminated vendor lock-in. Given it's a software-defined environment, we can now swap software components in and out at the click of an orchestrator. And that protects the service provider from underperformance from their vendors.

This is really happening because of a concept called disaggregation. This happens where the previous hardware-based environment is being split into a layered architecture, where at the bottom layer you have off-the-shelf hardware platforms such as those from Dell and HPE.

At the middle layer, you have an infrastructure component that is fully virtualized from companies like Wind River. And then at the top layer, you have an application layer where the telecommunications function is deployed. This enables competition at every layer. If the hardware vendor disappoints the customer, they can be swapped out for another. The same with the infrastructure and at the application layer.

This has really transformed the business that the service providers engaged in and given them significant leverage over their vendors. As we look at this, and Wind River made investments years ago to participate in this market with Studio Operator, we provide a differentiated cloud technology that is now used by Verizon in the US to deploy 5G and then Vodafone in the UK to deploy the very first open RAN network that's been deployed globally with Elisa in Finland, providing the fully automated infrastructure for deploying edge data centers and 5G networks, as well as companies like KDDI and NTT Docomo in Japan.

This is the first evolution of fully virtualized 5G Open RAN, and we're very excited about what this is doing for the industry because it's leveraging everything about Studio automation, orchestration, machine learning analytics, and distributed Kubernetes technology to host all these applications in the service provider market.

So as you can see, there's been significant transformations across the telecommunications and aerospace industries. And we find a really exciting software-defined future across all the verticals that we feed into. For further details on how this is happening, I'm going to introduce Avijit, who's our Chief Product Officer for the next segment.

ENABLING THE SOFTWARE-DEFINE VEHICLE

Avijit Sinha | Wind River's Chief Product Officer

Thank you Paul, and hello everyone. I am glad to be here to talk with you about how Aptiv and Wind River are enabling the software-defined vehicle.

But before I get started, we'd like to share a video that demonstrates the significant advantages of Aptiv and Wind River's joint capabilities for automotive OEMs looking to adopt next generation software architecture that is required for software defined vehicles. Let's get started...

[Wind River Video]

The investments that Aptiv has made in its business have very uniquely positioned it as the only full system solution provider. With industry leading advanced technologies that span not only every horizontal layer of the technology stack, but also all of the vertical domains.

Over the past five years, Aptiv has launched over 60 software programs with more than 20 customers, including:

A scalable, hands-free ADAS solution up to Level 2+ for a large global OEM, establishing Aptiv as a first technology provider to successfully deploy a scalable L2+ platform that is now in production.

The first infotainment solution with native Google Automotive Services for Volvo and Polestar and a best-in-class driver monitoring solution for multiple OEM systems, including GM's Super Cruise.

Our competitors are focused on a single layer of the tech stack or a single domain. Aptiv is uniquely positioned to operate across the full stack. Our full systems insights allow us to understand the interoperability between and synergies across technologies and domains which enable us to bring them together to create innovative, optimized solutions for our customers.

Through the commercialization of Smart Vehicle Architecture™, Aptiv is transitioning hardware-defined, legacy architecture to software-defined modern vehicle architecture. However, the vision of software-defined vehicles cannot be achieved unless our industry fundamentally changes the current approach to software development and lifecycle management.

Having executed more than 60 software programs with 20 different OEMs, we have experienced, first-hand, many software-related challenges OEMs face today.

The chart on the left represents the software release cadence and cost profile for developing, launching and supporting a highly complex software system over multiple generations, based on actual OEM data.

As you can clearly see, the costs increase exponentially over time with the introduction of each subsequent new generation. Meanwhile, high software costs do not translate into accelerated innovations – OEMs have struggled to update their software in a timely manner, managing to roll out minor updates only about once per year. Major updates can take a few years to implement.

A number of factors contribute to these challenges. First, across all vehicle generations, software was written on top of legacy distributed architecture with single-purpose ECUs, which lacks the scalability to support the increasing number of new features. Software was also tightly coupled to that underlying hardware, forcing OEMs to re-write the software code in each new generation whenever a hardware upgrade takes place.

Second, application software is built in a legacy, monolithic architecture. Software was written as a large, rigid block of code and was laced with interdependencies so complex that engineers find it easier to rewrite the entire code base than updating a specific component. This led to significantly higher development costs and slower release speed.

And third, all of this was exacerbated by the lack of modern software development and lifecycle management tools. During the development phase, engineers had to follow a disjointed and manual workflow, due to a lack of integrated toolchain. Post-production, there was simply no solution available that would allow OEMs to update software over-the-air.

These three key challenges led to lost revenue opportunities and higher development costs.

Of course, OEMs were very well aware of these challenges and started to look for solutions. However, due to a lack of off-the-shelf, readily-available third-party solutions, they had to resort to themselves by investing heavily in home-grown solutions. In addition to being a heavy undertaking in itself, such effort further compounded the challenges, as these home-grown solutions were often proprietary and lacked the ability to scale.

Aptiv and Wind River are here to change that. Aptiv and Wind River together provide a comprehensive product offering that can address all three challenges faced by OEMs today.

Wind River Studio offers the industry's first end-to-end platform to streamline software development, deployment and operations over the vehicle's lifecycle. Aptiv's containerized software and Wind River's edge software enable abstraction of software from hardware offering flexibility across semiconductor options and modular updates that reduce the cost of OTA.

All of this can be combined with Aptiv's SVA™ to develop, deploy and operate a next-generation, cloud native architecture for software defined vehicles. Our readily-available off-the-shelf solutions will allow OEMs to focus on truly brand differentiating effort, unlocking additional revenue opportunities while saving costs. Let us talk about each of these in more detail.

First, let's talk about the hardware architecture. As the automotive industry transitions toward software-defined vehicles, automotive OEMs are gradually rolling out their own version of the next-generation hardware architecture to support this vision. We expect nearly one of every four vehicles produced in 2030 to have a hardware architecture that is built on up-integrated, virtualized compute, and an optimized data and power distribution system.

This trend will create an addressable market of more than \$25 billion for advanced compute, in-vehicle operating systems and middleware and corresponding Dev-Ops platforms.

Aptiv foresaw this trend and has not only been a pioneer, but also a contributor to this trend by introducing our own Smart Vehicle Architecture™ years ago. We have gained meaningful market traction since then, and to date, we have booked \$10 billion in SVA™ awards, related to high performance compute.

Now, let's take a look at what is required for the software architecture.

As mentioned earlier, automotive software has historically been written as one monolithic block of code, with significant dependencies between components. Because of that, code changes and updates are often very costly and difficult.

To demonstrate this, let's try updating one feature, Adaptive Cruise Control or ACC. Changes in the ACC code require corresponding updates in the two libraries referenced by ACC, to ensure the overall software runs properly. The changes do not stop here. Because of the dependencies, change in one single code component leads to cascading changes throughout the entire code base, resulting in significant engineering effort.

This challenge can be resolved by containerization. With containers, software code can be modularized and packaged into independent, self-contained components. This allows isolated updates of code and reduces development time and complexity.

It is important to note that for safety critical software – like an ADAS feature – the benefits of containerization can only be realized through a RTOS that supports containerization at high levels of safety. This is to ensure that containerized software runs in a real-time, deterministic manner.

VxWorks is the only commercially available RTOS that has such capability.

Going back to our example of ACC update, now you see that the ACC feature can be updated without affecting other software components. This greatly simplifies software development, testing and integration.

Within each container, the software code can be further modularized – the software feature can be further broken down into individual services that communicate with each other via APIs, by implementing a modern microservices architecture. As software architecture further advances, even these individual services can be containerized. This will further improve software updatability and enable more dynamic management of applications running inside a vehicle.

And lastly, Wind River Studio offers a cloud-native, fully integrated DevSecOps platform to streamline all of an OEMs' software lifecycle management effort.

Historically, OEMs have used legacy toolchains that inorganically stitch together a set of third-party tools with no standard integration methods. This lack of standardization and integration led to significant manual effort and lost efficiency.

Wind River Studio instead is a customizable and integrated toolchain that enables automation and full traceability, which in turn contributes to faster development and reduced cost. Its capability to continuously certify software updates further expedites the development process by simplifying the traditional lengthy and costly safety certification process.

Similar benefits apply to the post-production phase. Wind River Studio provides a fully integrated environment for OEMs to continuously improve software and deploy updates over-the-air. This leads to better software features, reduced costs, improved customer experience and new revenue upside.

Having discussed the three solutions separately, let's now take a step back and look at how vehicles with next-generation hardware and cloud-native software will function in the world of the intelligent edge.

With the support of Helix and VxWorks, applications will be running in containers and can be easily scaled up and down across sub-systems. This enables more efficient use of compute resources. VxWorks also ensures that safety-critical applications still run safely and deterministically, as they are being scaled.

This capability to scale leverages Kubernetes, the same technology that hyperscale cloud service providers use to roll out highly available cloud services globally. With Kubernetes, software can also be orchestrated – not just within one vehicle – but across a fleet of vehicles, enabling software management at a large scale.

As software runs in the vehicles, data is generated and sent through the 5G network to regional data centers and then to the cloud. This connected infrastructure enables data to flow from the vehicles to Wind River Studio via the Digital Feedback Loop.

This data is then used to create Digital Twins of vehicles in the cloud for simulation. This enables OEMs to proactively discover and fix software issues, driving down costs of software development and defects. It will also allow OEMs to better understand consumer behaviors and unlock additional value by building tailored features that improve consumer experience. Once vehicle data is analyzed in Wind River Studio, developers can update the software code and deploy it to vehicles. This enables a single plane of glass experience across development, deployment and operations, all within Wind River Studio.

So we just walked through the automotive technology stack that is required to enable the software-defined vehicle and how Aptiv and Wind River's combined capabilities make us a partner-of-choice to OEM customers.

Now, I'd like to hand it off to Anant Thaker, Aptiv's Chief Strategy Officer and Senior Vice President Product, who will dive deeper into the value that we unlock for customers and our commercial progress.

ENABLING THE SOFTWARE-DEFINE VEHICLE

Anant Thaker | Aptiv's Strategy Officer and Senior Vice President of Product

Thanks Avijit, and hello everyone!

Earlier, Avijit showed you the chart on the left, which represents a legacy approach to software development and deployment in which software reuse is limited, updates are difficult and infrequent and costs compound significantly over time.

These challenges can be addressed with the modern, cloud-native approach enabled by Aptiv and Wind River's solutions, with the results shown on the right hand side.

As you can see, software development, integration and testing is streamlined, with faster time to market and lower up-front costs, and true lifecycle management is enabled, allowing for: More frequent software releases, with new and improved features, and reduced spending on warranty and recalls, as issues can be detected and addressed with ongoing software updates.

Based on Aptiv's experience, we estimate these solutions can ultimately reduce OEM software costs of development, deployment and post-production operations by about 25%. They also support new OEM business models by enabling more frequent feature releases and updates over a vehicle's lifecycle.

Our solutions empower OEMs with greater flexibility in other key layers of the technology stack, including two of the largest and fastest-growing spend categories: semiconductors and cloud.

Wind River's in-vehicle software platform is supported by a broad range of SoCs, enabling software applications to be abstracted from the underlying hardware. This allows greater portability of software across vehicle platforms, and provides OEMs with more choices among SoCs to optimize system performance and cost, while also improving supply chain resiliency.

And through compatibility with major cloud platforms, Wind River Studio can run on an OEM's cloud of choice. By decoupling the development environment, toolchain and data platform from the underlying cloud infrastructure, this again drives greater flexibility.

As a result, Aptiv and Wind River solutions enable OEMs to accelerate innovation, optimize performance and improve scalability, while also avoiding vendor lock-in and managing costs throughout the technology stack.

In addition to providing flexibility across the technology stack, our software products themselves are modular in nature, allowing customers to pick and choose the solutions that best fit their needs.

For customers that are looking for a full-system solution for software-defined vehicles, Aptiv can offer a cross-domain package of hardware and software products optimized at a system level. This includes our SVA™ hardware, Wind River's in-vehicle software platform and Aptiv's application software, with end-to-end lifecycle management enabled through Wind River Studio.

For OEMs aiming to develop their own applications, in ADAS or infotainment for instance, we can offer hardware, Wind River's software platform, differentiated building blocks such as Aptiv's perception stack, as well as Wind River Studio to enable these customers to develop, deploy and manage software more seamlessly and at lower cost.

Finally, based on how we have architected our software offerings, we are increasingly positioned to sell software as standalone products that can be integrated with hardware from other providers.

And across any of these models, we are able to leverage Aptiv's deep experience in integration, testing and validation to ensure execution.

This go-to-market approach is differentiated from several of our competitors who typically propose rigid, "black box," vertically integrated offerings and lack the cross-domain portfolio and experience to offer integrated solutions.

To illustrate how unique our portfolio and go-to-market model is, let's take our Gen 6 ADAS platform as an example.

We've purposefully architected this platform to be open and modular and capable of meeting the needs of all OEMs in the market. Starting with the bottom of the stack:

We provide an advanced sensor suite and high performance compute solutions. When integrated with Wind River's software platform, higher-level software is abstracted from the underlying hardware, which reduces OEM dependency on any particular SoC.

One layer up the stack, our AI/ML-based radar software can be paired with multiple computer vision solutions to provide best-in-class, high availability perception to enable highly automated ADAS features.

OEMs can then leverage the Wind River software platform, our perception stack and Wind River Studio to build, test, deploy and update their own in-house features. Or they can opt for a full-system solution, leveraging Aptiv's market-leading ADAS features, which use service-based architecture and containers to streamline updatability, while maintaining the highest levels of safety certification.

As we bring Aptiv and Wind River solutions to our customers, we are also building a robust, open and collaborative ecosystem of partners that complement our technology.

We are proud to work with some of the leading silicon and cloud infrastructure providers in the market. For example, Wind River is partnering with Samsung to provide customers with this technology, including Helix, VxWorks, and Linux, on Samsung's Exynos platform, enabling advanced software applications across all levels of safety, including in infotainment, cabin monitoring, and ADAS.

And through Wind River's collaboration with Horizon Robotics, OEMs can leverage a fully integrated ADAS solution tailored to address the unique needs of the Chinese market with Horizon's Journey series and Wind River's full edge-to-cloud portfolio.

Finally, Wind River Studio can run on an OEM's cloud of choice and supports both Amazon Web Services and Microsoft Azure, two of the most widely adopted infrastructure providers in the automotive industry.

Bringing differentiated solutions to meet our customers' needs has resulted in strong commercial success for Aptiv and Wind River, and the flexible go-to-market approach I discussed earlier is reflected in our awards, which span multiple products, platforms and regions, and range from providing a full-stack ADAS system for Geely, with Aptiv's perception and L2+ features integrated with the Wind River software platform, to enabling Hyundai Mobis and AutoEver's development of next-generation connected vehicles using Wind River Studio.

These commercial wins underscore the value of our portfolio, and the robust engagement we continue to have with several OEMs on enabling their journey to software-defined vehicles indicates that this is only the beginning.

So when you step back, we see that the emergence of the intelligent edge and the shift to a software-defined future presents significant growth opportunities for Aptiv and Wind River, as Benjamin highlighted earlier. Our combined portfolio can address a market size of \$13 billion today, growing rapidly to roughly \$37 billion by 2030.

And building on the strong momentum and market traction we are seeing for our solutions across industries, we aim to raise our software revenue from \$400 million today to \$3 billion in 2030, at a growth rate that is double that of the market.

To summarize, the transition to a software-defined future and the demand for an intelligent edge is happening now across multiple, safety critical industries.

Wind River has a proven track record of bringing cloud-native solutions to the edge to address some of the most demanding, real-time applications from aerospace and defense to telecommunications.

And Aptiv and Wind River are now uniquely positioned to leverage these same capabilities to address the needs of the automotive industry by modernizing software architecture as well as the development environment and lifecycle management.

As the industry faces a pivotal moment to significantly change how software is developed, deployed and managed to truly unlock software-defined vehicles, we remain the partner of choice for OEMs embarking on this transformative journey.

Thank you for your time, and now I'd like to hand it over to Jane to moderate us through Q&A, where we'd be happy to address any questions you might have.

LIVE Q&A SESSION

Jane Wu | Vice President, Investor Relations and Corporate Development

We hope that you enjoyed the presentation. We will now begin our Q&A session with our presenters. And as a reminder, you can submit questions in the Slido text box at the bottom of your screen.

So first question, how are customer engagements going? And are OEMs recognizing the value of transitioning to modern software architectures?

Kevin Clark | Chairman and Chief Executive Officer

I'll start off and then I'll turn it over to my colleague Avijit. I'd break that question up into two pieces.

I think when you look at the telecommunications, the aerospace and defense industries, those places where Wind River's been involved for a number of years -- their primary industries -- those customers have recognized the importance of virtualizing hardware quite some time ago. And as a result of that, the benefits of being software defined and the importance of how you architect software to get the full flexibility, the full cost reduction, the full optimization that they're looking for.

From an automotive standpoint, which we're bringing Wind River into, I would say, Aptiv's experience from a hardware standpoint, from redefining the hardware, Smart Vehicle Architecture™, which we've been talking about for a number of years, is intended to enable the software defined vehicle. Obviously, we've made a lot of progress and have a lot of traction with customers.

From a software architecture standpoint, I'd say today, customers fully recognize the importance of having a software defined vehicle. I think there's some element of still struggling with the implementation associated with that. Given some of the struggles the industry's had with programs and software development over the last couple of years, I'd say there has been a lot of progress in terms of understanding the importance of software architecture and the role that it plays.

We've had a number or have a number of OEM engagements today, so I would expect a tremendous amount of momentum as we go forward. But I'll turn it over to Avijit to be a bit more granular about some of our discussions.

Avijit Sinha | Wind River's Chief Product Officer

Thank you, Kevin. Our customers across industries -- aerospace and defense, as you mentioned, telecommunications and automotive -- are embracing cloud native technologies and software development business models to bring differentiated solutions to their customers. But as Benjamin referenced in his presentation, customers are on a maturity cycle. There are certain customers in certain industries who've been early in adopting these technology patterns and others who are coming to it now.

So in the aerospace and defense industry, as Paul mentioned in his presentation, they've embraced virtualization as a means to get flexibility across silicon choices. So there's been early adoption there, and more recently in telecommunications we've been working with our partners such as Intel and Samsung and others to give them the ability to adopt virtualization as they virtualize the RAN in the network.

And more recently with Aptiv, we're seeing traction in the automotive industry for our technologies across edge and studio as they seek to bring software defined architecture into the vehicle and bring life cycle management capabilities into the cloud. So we're seeing tremendous adoption in aerospace and defense and telecommunication and significant interest now with Aptiv in the automotive industry, as that industry starts to embrace these technology patterns and drive innovation in that industry.

Jane Wu | Vice President, Investor Relations and Corporate Development

Next question. Can you dive deeper into what software Aptiv and Wind River are doing in your ADAS solutions? How will Wind River be integrated into Gen 6 ADAS?

Anant Thaker | Chief Strategy Officer and Senior Vice President, Product

Yeah, maybe I can start there. As we mentioned in the presentation, our Gen 6 ADAS platform it's built off of the Wind River software platform. So, we're running Wind River VxWorks, we're running Helix Hypervisor, we're running Linux in some instances. And one of the things we highlighted and what really attracted us to Wind River that made it so differentiated was that VxWorks is the only real time operating system that can have the highest levels of AS-IL certification while also bringing cloud native technologies into that environment. So it's the only RTOS that supports containers at an application level. And our ADAS stack is a service based architecture that takes advantage of containers, so we're excited to integrate these two together. And we're going to be demonstrating that at CES. So our ADAS stack is built off of and will be leveraging the Wind River software platform. And then our engineers who

are writing those applications and deploying and managing the vehicles will be doing so using Wind River Studio.

Jane Wu | Vice President, Investor Relations and Corporate Development

Great. Third question. Does Wind River complete your go-to-market capability? Or are there other areas you need to develop internally and or buy in?

Anant Thaker | Chief Strategy Officer and Senior Vice President, Product

So I can start. I think that there's always opportunities to broaden our capabilities and broaden where we participate in the stack. But we do think that now it gives us, as we mentioned in the prepared presentation, such a flexible go-to-market business model where we can provide our customers with the full hardware architecture as well as the core software platform on top of which we can build applications -- So that's our SVA™ Hardware as well as the Wind River software platform. And then we can offer Studio for customers to develop those applications and manage the fleets post production. And then where we have really differentiated applications at a domain level-- it could be at ADAS, it could be in cabin sensing, it could be across the other parts of the vehicle. We can also offer those applications to build full system solutions. So, we have a complete offering for really picking flexible, modular, and full system solutions to our customers.

Jane Wu | Vice President, Investor Relations and Corporate Development

We have a question on unit economics and how does Wind River charge for Studio versus Edge software? Is it a subscription per seat?

Avijit Sinha | Wind River's Chief Product Officer

Yeah. So if you look at our products in the customer segments we serve, our Studio developer product is geared towards developers. Our Studio operator product is geared towards operations and lifecycle management. And our Edge software is geared towards software in the asset itself, whether it's a plane or a telecom tower or a vehicle.

So when you look across these products and the capabilities they provide on the developer side, it's a per-seat based business model. On the operation side, it's a subscription based model and licensing of the software that goes into the vehicle or the asset.

And then when you look at the lifecycle management, we have shifted our customer base from licensing to subscription based business model because that aligns with how their cost structure is over the life cycle of the asset -- that they're trying to manage and it gives us a good way of having recurring revenue over the engagement we have with our customers.

Jane Wu | Vice President, Investor Relations and Corporate Development

The software defined, cloud enabled vision seems very dependent on OEMs that all want to have tight control over consumer modernization. Does this help or hurt Wind River and or Aptiv?

Kevin Clark | Chairman and Chief Executive Officer

Maybe I'll start, if that's okay. As you saw in the presentations, and I think it's reflected in some of the answers to the questions we've given, we're really about how do we enable our customers to create a safe, secure business model that allows for recurring revenues. That is our objective.

Aptiv and Wind River are not B2C companies, they're B2B companies. To the extent we can enable a recurring revenue stream for our customers, the reality is the markets we serve actually get bigger. You know, the ADAS market, the User Experience market, in similar situations in the telco, A&D and other markets. So I wouldn't say it has any impact on how we go to market or our business model. It's something that we're really trying to drive, so it actually increases the size of the markets that we actually operate in.

Anant Thaker | Chief Strategy Officer and Senior Vice President, Product

Can I just add one thing? I think that the question was around, is it reliant on how OEMs control or affect monetization on the revenue side? A big driver for us for moving to cloud native architectures and vehicles is also to help with the cost reduction, to develop software faster and to reduce the cost of delivering it, which is not at all dependent on how it's ultimately monetized. So we think this enables OEMs to develop software faster at a lower cost and deployed and manage even issues and updates and warranties, over the full lifecycle of the vehicle.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. So just in terms of the competitive landscape, what is it by industry by product?

Avijit Sinha | Wind River's Chief Product Officer

In the aerospace and defense industry, as it pertains to software that goes in the assets, we have competitors like Green Hills. In telecommunications as we brought virtualization technology, competitors are like Red Hat and VMware. In the automotive industry, obviously, QNX in the vehicle.

But when you look across the spectrum of these industries and edge software and cloud native capabilities, none of these players are able to offer an end to end solution. So we offer a very differentiated set of capabilities that span not just the asset, but the cloud capabilities that are essential to drive the lifecycle management of these assets.

So we bring a very differentiated portfolio to the market, and that's why the interest that we see across aerospace, telecommunications, and automotive for our products.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. So auto OEMs are building out their own software capabilities. Do you see risk of insourcing?

Kevin Clark | Chairman and Chief Executive Officer

Maybe I'll take a shot if that's okay. The amount of software going into vehicles is significant. When you think about ADAS, User Experience, electrification, enabling all that in an efficient way, it requires incremental software. So the market is significant, the growth opportunity is significant.

A number of OEMs are interested in participating in that, from an economic standpoint, cost standpoint, and also from a brand standpoint. The reality though is taking the old model of developing and integrating software and insourcing that into an OEM does nothing but create additional problems. And again, you've heard about those problems, those challenges.

Our objective is how do we create a very efficient, flexible, low cost process for our customers to have choice to enable more software to go into the car, which requires software to be completely rearchitected for the automotive industry relative to where it's been historically. And to put them in a position where they can create recurring revenue models, as they've all talked about. And if you look at historically how software was developed and integrated in the car, clearly there was a hardware issue, but there also was a software architecture issue. And our objective is really to solve that particular problem to give the OEMs flexibility, whether it's they want to develop some of the software and integrate it or they want to rely on players like ourselves to do some or all.

I don't know if, if anyone wants to add to that.

Paul Miller | Chief Technology Officer

One of the key technology components that enables that, we've seen that from the work that we've done in aerospace and defense, as well as telecommunications, is the desire to have faster time to market, to be able to bring software functions into the vehicle or the application environment that enables faster monetization and new services. And the legacy software architectures were very monolithic, and it required you to deploy the entire system, and therefore updates in the field and the ability to deploy new applications were really restricted. And so moving to these more modern cloud native approaches enables a customer to monetize much more rapidly, bring more applications and really participate in the software value ecosystem.

Benjamin Lyon | Chief Technology Officer and Senior Vice President

Yeah, just to layer one thing onto that. This is part of what was so interesting to Aptiv about the Wind River platform is it's an open platform, and that is what provides that flexibility really to our customers.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. All right. Next question. Who will own the data that sits in Wind River Studio? Will the OEMs own this data? Do you anticipate recurring revenue models or do OEMs have appetite for it?

Avijit Sinha | Wind River's Chief Product Officer

Yes, fundamentally our value proposition is that we enable our customers to adopt our technology to build their business and their set of services that they provide to their customers. So the data that's emanating from their assets and their customers will come into Studio, but it's owned by our customers. We give them the capabilities to take that data, to drive analytics on it, and to derive insights. And in the

future, using AI and machine learning, to take that data and build AI models that can help drive differentiated offerings to their customers. So the data is theirs. We provide them the capabilities to harness the value out of the data.

Jane Wu | Vice President, Investor Relations and Corporate Development

Can you provide more insight for the build-up for the 25% software program savings? Where will these savings come from?

Anant Thaker | Aptiv's Strategy Officer and Senior Vice President of Product

I can start there. So where that number came from really was we looked at our own experience working with a number of OEMs on the cost of software development. So everything from the initiation of a program to when we're actually launching the product and then deploying that software. So the cost of updating over-the-air with a full lifecycle of a number of fleets. And then the third bucket was operating those vehicles. So managing lifecycle issues, warranties, recalls and other things like that.

And when we looked at it, we looked at the total cost, for an example, an OEM, a large scale software program, it could be north of \$2 billion dollars over a ten year lifecycle for say ten million vehicles. And when we looked at how Wind River solutions could actually affect change there, speed development, drive down cost development on the deploy side because we're shifting to containerization, it means the updates can be smaller, it can reduce the cost of that OTA updating, and then because you can mitigate issues in the field, you can reduce the cost of warranties and lifecycle management.

And so when we added that all up, we said that you found that it was around a 25% cost savings. And we're confident we can bring that to our customers.

Jane Wu | Vice President, Investor Relations and Corporate Development

There are other industry leaders in operating systems for automotive. How does Wind River compare and how will you win share?

Avijit Sinha | Wind River's Chief Product Officer

There are obviously incumbents who provided that capability to the automotive industry for years. However, many of these competitors are more anchored in the legacy way of software development and operating system constructs even.

Our, as Anant mentioned in this presentation, VxWorks with support for containers, and the fact that it's certified, offers a more modern software that's required for the distributed architecture in the vehicles and then complemented with cloud native DevSecOps capabilities, the ability to go from edge to cloud. These are things that are net new for the industry. They anchor them more into the future as opposed to existing incumbents who are anchored in the past.

So we've got a very full set of portfolio capabilities that bring customers forward into the future as they seek to embrace software driven, defined vehicle models. So I think we are very well positioned to win in the industry there.

Kevin Clark | Chairman and Chief Executive Officer

And if I can add, none of those competitors have those same capabilities, either architecture or DevSecOps tools, that Wind River has and has been implementing in other industries very successfully.

Jane Wu | Vice President, Investor Relations and Corporate Development

Sounds good, thanks. So in terms of the risks to the forecast and opportunities, what do you think are the big risks in achieving your plan?

Kevin Clark | Chairman and Chief Executive Officer

I'll start. I think our plan across industries is actually very prudent and very reasonable. As we've been saying for the last couple of years, our customers in the automotive industry are struggling and are looking for help in enablement. There's a great deal of interest in what Wind River has to offer.

When you look at telco, A&D and other markets, the reality is the understanding of the need is pretty well penetrated across the customer base globally. So there's a tremendous amount of pull, both within existing customers as well as with other customers. So the momentum is very, very strong. So we have a high level of confidence in the forecast that we've submitted.

Jane Wu | Vice President, Investor Relations and Corporate Development

Okay, thanks. Are there opportunities for further tuck-in software acquisitions to leverage Wind River core competencies? What could some of those opportunities be?

Kevin Clark | Chairman and Chief Executive Officer

Avijit or Paul, do one of you want to talk about it?

Paul Miller | Wind River's Chief Technology Officer

I think that as we look at our portfolio now, we've come through several years of investment and acquisition that we think has brought together all the assets that we really need to satisfy our customer demands.

There's always going to be opportunities to add new capabilities to the platform, but I think we today have the fundamental assets that we need for the entire software lifecycle management. As you saw in the presentation, not only the DevSecOps tool chain, the infrastructure, very modern capabilities with cloud native assets and then operational capabilities to manage those deployed systems. So a very rich set of capabilities that's taken some time to get to.

So we consider ourselves pretty complete at this point, but we'll always look for new opportunities where it makes sense to add new software capabilities to our portfolio.

Jane Wu | Vice President, Investor Relations and Corporate Development

How do you use AI/ML today with Wind River or in your ADAS solutions, or how do you plan to incorporate it in the future?

Benjamin Lyon | Senior Vice President and Chief Technology Officer

Sure. Well, it's interesting. Aptiv has actually been investing in AI for quite some time, and you can see that in our sixth generation ADAS, both on the perception side, but also in the driving policy, so how the vehicle makes decisions. So we see that as something that will continue to lead and drive forward.

In addition, we look into the back office functions for how we manage our business. And we look at what are the opportunities for using AI to identify patterns, anomalies, and figure out ways to be more efficient.

And then lastly, outside of automotive, if you think about telco and A&D where you're managing a large distributed system, we're looking at opportunities for applying AI to how you manage that entire set of assets in order to detect issues in the system and either elevate those issues for resolution by an operator or automatically take care of them.

Kevin Clark | Chairman and Chief Executive Officer

Maybe, Paul, you could layer on to that.

Paul Miller | Wind River's Chief Technology Officer

Yeah so in Wind River, we have four key pillars to our AI strategy.

The first is that within the DevSecOps tool chain, we want to enable software developers to use AI capabilities, whether it be AI automated test or code completion capabilities, that help them further reduce the cost of developing software and improve its predictability in time to market.

The second element is we really want our customers to be able to build AI and ML applications. So a set of libraries and assets that they have that help them build their own solutions as they build their own products.

The third component is within our Edge portfolio, the operating systems and infrastructure that we provide. We need the ability to host machine learning applications in particular on the edge as well as AI functions. And so things like TensorFlow Lite support and capabilities that we need to host those applications are that part of the strategy.

And then finally, as Benjamin just mentioned, in aerospace and defense and automotive, and in telecommunications in particular, the operational capabilities of AI are quite stunning. The ability to do predictive outage avoidance or predictive analytics, event correlation, root cause analysis, these turn weeks of debugging in the field into minutes. And so leveraging the technologies across those four pillars is how we look at investing in AI.

Jane Wu | Vice President, Investor Relations and Corporate Development

Sounds exciting. How would Wind River help OEMs decouple from underlying hardware and why is this important?

Anant Thaker | Aptiv's Strategy Officer and Senior Vice President of Product

Maybe I can start there. So I'll start with why it's important.

I think what we've seen in our experience is as we're moving to more SVA™-like architectures, and we shared it in the presentation, that one of the largest costs for OEMs is the underlying silicon and SOCs. And so by driving some more flexibility and choice, which the Wind River software platform enables, that allow customers to take their existing software stack and potentially migrate it to a different SOC platform and pick and choose what's best for them for a specific application. We think that gives them flexibility, but then given the nature of how costly it is as a part of the total vehicle architecture, it can really help them optimize on cost as well.

Avijit Sinha | Wind River's Chief Product Officer

As Anant said, being a fundamental platform provider, we appreciate the value of an ecosystem and partnerships. And so as Anant mentioned in his presentation, the two biggest cost drivers for our customers in automotive are cloud costs and silicon costs.

Anant already touched upon the value that can be accrued by having diversity across silicon choices. So we have a number of partnerships in Wind River across silicon with Samsung, with Ambarella, with NXP, with Texas Instruments and many others in the pipeline there. So we're going to enable a lot of flexibility and choice for our OEM customers there, so drive down costs there. And on the cloud side too, our Studio capabilities run on AWS and Azure that gives them flexibility. So having flexibility across both of these cost drivers gives our customers the ability to not be vendor locked-in and have long term flexibility in their business model.

Jane Wu | Vice President, Investor Relations and Corporate Development

Thank you. This is our final question. Is it conceivable that other tier-1 suppliers will use Wind River for ADAS programs despite Aptiv not being the tier-1 provider?

Anant Thaker | Aptiv's Strategy Officer and Senior Vice President of Product

I'll start. So I'd say absolutely. By design, right, we want it to be flexible, we want others to work in the solution. And I would say that one of the core elements of Studio Developer, in particular, is that we're integrating a lot of third party tools. And some of those third party tools come from our competitors or they come from others in the ecosystem. So we absolutely want to be able to enable the broader industry to use it. And it's not just about Aptiv or our customers using the product.

Kevin Clark | Chairman and Chief Executive Officer

Yeah, I'd say one of the rationales as it related to Wind River-Aptiv transaction was obviously how can we help Wind River grow in the markets that they currently operate? How do we bring that capability into the automotive market where we see a huge need? But also how do we enable Wind River to grow? And our objective is to make sure Wind River has the strongest solution.

And again, we're about choice and flexibility to our customers. Whether our customers want to deal directly with Wind River, and we have situations where that's occurring today, or they want to deal with Wind River through another tier-1. That's absolutely fine. That's something that we encourage and we're incentivizing the Wind River team to do.

Jane Wu | Vice President, Investor Relations and Corporate Development

Sounds good. Alright, any final comments for the team?

Kevin Clark | Chairman and Chief Executive Officer

Thank you for your time. As you can tell, we're very excited about Wind River, the addition of Wind River to the Aptiv overall portfolio of technologies. We tend to talk a lot about what's going on in the automotive industry based on our perspective and where we come from.

And bringing Wind River into automotive, we're convinced, solves a lot of problems for our OEM customers and presents tremendous opportunities for Aptiv.

However, there are also tremendous untapped opportunities in telecommunications and aerospace and defense and the industrial markets that the Wind River team is also very focused on. And we're really excited about what Wind River brings not only into the automotive market, but into those markets as well.

Jane Wu | Vice President, Investor Relations and Corporate Development

Great. Thank you everyone. And thank you very much for joining.

Kevin Clark | Chairman and Chief Executive Officer

Thank you.